Quantum Computation with Quantum Dots: Decoherence and Fidelity Limits

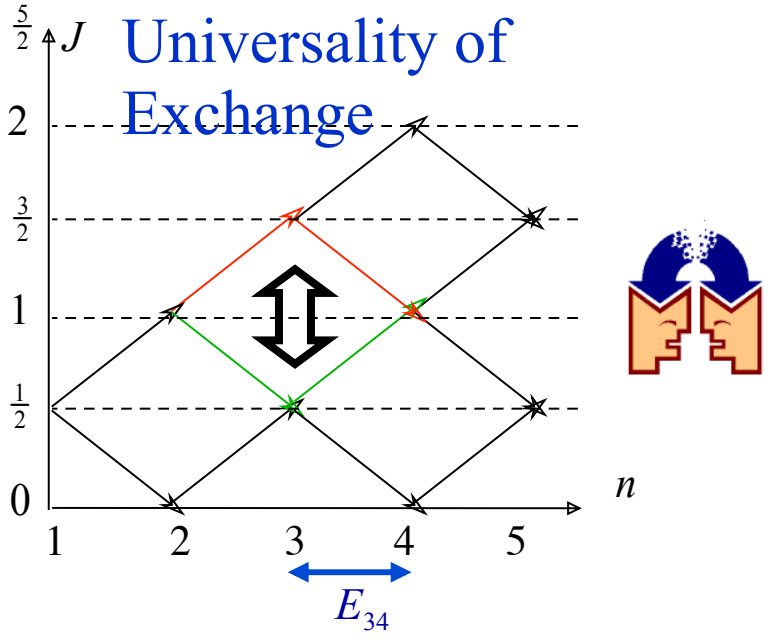
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ARDA

Objectives

- Overcoming decoherence limitations to scalable quantum computing realizations
- Decoherence prevention, error correction, and fault tolerant computation
- Development and simulation of realizable models for QC employing quantum dots

Universality of Exchange



Objective Approach

- Control of decoherence via Subspace or Subsystem selection (DFSs)
- Universal, fault tolerant QC on DFS
- Optimal design of quantum dot implementations
- Numerical Simulation of QC models

Status

- Universal QC achieved using 2-body exchange interaction only
- Fault tolerant QC demonstrated on DFSs
- Explicit gate sequences for some DFSs
- Proposal for QC on quantum dots connected with linear support

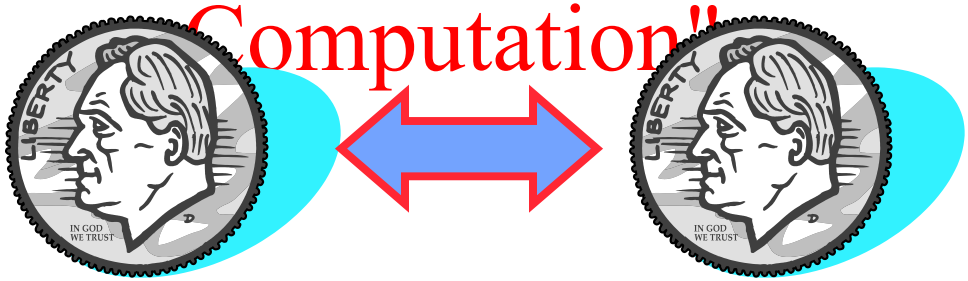
Research Plan

Quantum Computing with Quantum Dots: Decoherence and Fidelity Limits

K. B. Whaley, UC Berkeley

- Progress on last year's objectives
  - DFS concatenated with QECCs shown to be stable under exchange errors
  - fault tolerant quantum computation demonstrated on decoherence free subspaces and subsystems (DFSs)
  - explicit gate sequences derived for QC on DFSs defined by: i) collective decoherence, and ii) multiple qubit errors
  - universal quantum computation shown to be achievable using 2-body exchange as the only physical interaction
  - low decoherence QC scheme using quantum dots connected with linear support
- Research plan for the next 12 months
  - search for other physical interactions possessing asymptotic universality
  - physical realizations of DFSs and computation thereon
  - stochastic wave function simulations: QC based on exchange interaction, QC on quantum dot arrays, minimizing decoherence via DFSs, …
  - investigation of spin coherence in quantum dots
- Long term objectives
  - overcome decoherence limitations to scalable quantum computation
  - characterize physical systems for fault tolerant, universal QC with DFS-QECCs
  - establish realistic scenarios for scalable qubit registers employing quantum dots

Quantum Computing on Decoherence-Free Subspaces
"Surprise: A New Paradigm For Quantum Computation"



Dave Bacon

University of California Berkeley

Departments of Physics and Chemistry

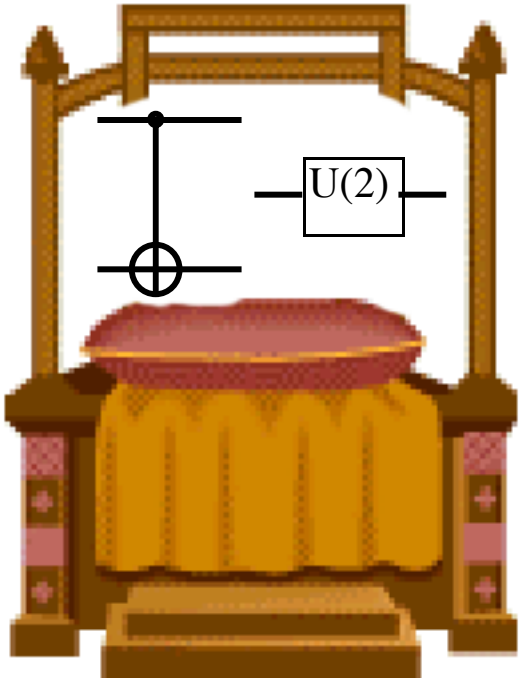
Cult of Universality

Universal Quantum Computer

≡ a quantum computer which can be programmed to perform any algorithmic manipulation on quantum information.

Set of Universal Quantum Gates

≡ a set of operations/gates which, acting on the quantum information, can be used to implement (to any desired accuracy) any unitary evolution of the quantum info.



The Royal King and Queen of Universal Quantum Gates

CNOT and 1-qubit rotations

Other methods (TAILOR TO AVAILABLE PHYSICS)???!!!





Collective Operations



ARDA

Single qubit angular momentum operators. We will think of a qubit as a spin-1/2 particle.

$$s_x = \tfrac{1}{2}\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \quad s_y = \tfrac{1}{2}\begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \quad s_z = \tfrac{1}{2}\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \qquad \left[s_\alpha , s_\beta \right] = i\varepsilon_{\alpha\beta\gamma} s_\gamma$$

Total angular momentum operators on n qubits → Collective su(2) operators:

$$\boxed{S_\alpha^n = \sum_{i=1}^{n} s_\alpha^{(i)}} \quad \text{where } s_\alpha^{(i)} = I \otimes \ldots \otimes I \otimes \underbrace{s_\alpha}_{i\text{th qubit}} \otimes I \otimes \ldots \otimes I \qquad \left[S_\alpha^n , S_\beta^n \right] = i\varepsilon_{\alpha\beta\gamma} S_\gamma^n$$

$$S_x^2 = \frac{1}{\sqrt{2}} \left(\begin{array}{ccc|c} 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ \hline 0 & 0 & 0 & 0 \end{array}\right) \qquad S_y^2 = \frac{1}{\sqrt{2}} \left(\begin{array}{ccc|c} 0 & -i & 0 & 0 \\ i & 0 & -i & 0 \\ 0 & i & 0 & 0 \\ \hline 0 & 0 & 0 & 0 \end{array}\right) \qquad S_z^2 = \left(\begin{array}{ccc|c} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ \hline 0 & 0 & 0 & 0 \end{array}\right)$$

triplet states $\begin{cases} |J=1, m_J=1\rangle = |00\rangle \\ |J=1, m_J=0\rangle = \frac{1}{\sqrt{2}}\left(|01\rangle + |10\rangle\right) \\ |J=1, m_J=-1\rangle = |11\rangle \end{cases}$ singlet state $\left\{ |J=0, m_J=0\rangle = \frac{1}{\sqrt{2}}\left(|01\rangle - |10\rangle\right) \right.$

Intro quantum mechanics: addition of angular momentum for two spin-1/2 particles.

- Two spin-1/2 can be combined into a spin-1 triplet and a spin-0 singlet.

Degeneracy

- spin-1 (triplet) plus spin-1/2 can produce either a spin-1/2 or a spin-3/2.
- spin-0 (singlet) plus spin-1/2 can produce spin 1/2.

$$S_x^3 = \left(\begin{array}{cccc|cc|cc} 0 & \sqrt{\frac{3}{2}} & 0 & 0 & 0 & 0 & 0 & 0 \\ \sqrt{\frac{3}{2}} & 0 & 2 & 0 & 0 & 0 & 0 & 0 \\ 0 & 2 & 0 & \sqrt{\frac{3}{2}} & 0 & 0 & 0 & 0 \\ 0 & 0 & \sqrt{\frac{3}{2}} & 0 & 0 & 0 & 0 & 0 \\ \hline 0 & 0 & 0 & 0 & 0 & \frac{1}{2} & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{1}{2} & 0 & 0 & 0 \\ \hdashline 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{2} \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{2} & 0 \end{array}\right)$$

$$\left\{\begin{array}{l} \left|J=\tfrac{3}{2}, m_J=\tfrac{3}{2}\right\rangle = \left|000\right\rangle \\ \left|J=\tfrac{3}{2}, m_J=\tfrac{1}{2}\right\rangle = \tfrac{1}{\sqrt{3}}\left(\left|001\right\rangle+\left|010\right\rangle+\left|100\right\rangle\right) \\ \left|J=\tfrac{3}{2}, m_J=-\tfrac{1}{2}\right\rangle = \tfrac{1}{\sqrt{3}}\left(\left|011\right\rangle+\left|101\right\rangle+\left|110\right\rangle\right) \\ \left|J=\tfrac{3}{2}, m_J=-\tfrac{3}{2}\right\rangle = \left|111\right\rangle \end{array}\right.$$

$$\left\{\begin{array}{l} \left|J=\tfrac{1}{2}, m_J=\tfrac{1}{2}\right\rangle = \tfrac{1}{\sqrt{2}}\left(\left|01\right\rangle-\left|10\right\rangle\right)\left|0\right\rangle \\ \left|J=\tfrac{1}{2}, m_J=-\tfrac{1}{2}\right\rangle = \tfrac{1}{\sqrt{2}}\left(\left|01\right\rangle-\left|10\right\rangle\right)\left|1\right\rangle \end{array}\right.$$

$$\left\{\begin{array}{l} \left|J=\tfrac{1}{2}, m_J=\tfrac{1}{2}\right\rangle = \sqrt{\tfrac{2}{3}}\left|00\right\rangle\left|1\right\rangle-\sqrt{\tfrac{1}{3}}\left\{\tfrac{1}{\sqrt{2}}\left(\left|01\right\rangle+\left|10\right\rangle\right)\right\}\left|0\right\rangle \\ \left|J=\tfrac{1}{2}, m_J=-\tfrac{1}{2}\right\rangle = \sqrt{\tfrac{2}{3}}\left|11\right\rangle\left|0\right\rangle-\sqrt{\tfrac{1}{3}}\left\{\tfrac{1}{\sqrt{2}}\left(\left|01\right\rangle+\left|10\right\rangle\right)\right\}\left|1\right\rangle \end{array}\right.$$

These spin-1/2 states come from combining a spin-0 two qubit state and a spin-1/2 single qubit state

These spin-1/2 states come from combining a spin-1 two qubit state and a spin-1/2 single qubit state

There is a **DEGENERACY** in the action of the 3 qubit collective su(2) operation!

Protection

S_x^3 over the *degenerate portion* $= \begin{pmatrix} 0 & \frac{1}{2} & 0 & 0 \\ \frac{1}{2} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{1}{2} \\ 0 & 0 & \frac{1}{2} & 0 \end{pmatrix} \rightarrow I_2 \otimes s_x^2$

$$S_\alpha^3 \rightarrow s_\alpha^4 \oplus \left(I_2 \otimes s_\alpha^2 \right)$$

$$S_\alpha^4 = \left(\begin{array}{c|cc} s_\alpha^4 & 0 & \\ \hline 0 & s_\alpha^2 & 0 \\ & 0 & s_\alpha^2 \end{array} \right) = \left(\begin{array}{c|c} s_\alpha^4 & 0 \\ \hline 0 & I_2 \otimes s_\alpha^2 \end{array} \right)$$

$$\rho = \left(\begin{array}{c|c} 0 & \\ \hline & \rho_S \otimes \sigma \end{array} \right)$$

-action of S_α^3 on subspace which has the degeneracy acts as identity on a tensor component!

-info in ρ_S **will NOT decohere**! info in σ will decohere! (think of $I \otimes \sigma_\alpha$ for two qubits)

(KLV) Emanuel Knill, Raymond Laflamme, Lorenza Viola, Phys. Rev. Lett. **84**, 2525 (2000)

Decoherence-Free Subsystems (KLV)

General system-bath Hamiltonian $H_{SB} = \sum_{\alpha} A_\alpha \otimes B_\alpha$

A_α: system operators; B_α: linearly independent bath operators

A_α form an algebra A (add, multiply them) which is decomposable into subsystems:

$$\mathrm{A} = \bigoplus_{J=1}^{r} \left(I_{n_J} \otimes M_{d_J} \right)$$

linear operators on a d_J dimensional space

I_d is the $d \times d$ identity operator

n_J is the degeneracy of the Jth irrep

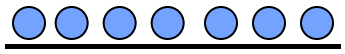
Quantum information encoded into a single degenerate component subsystem will not decohere! Decoherence-Free Subsystems (DFSs)

When $d_J = 1$ the information will be encoded over a subspace: Decoherence-Free Subspaces

"The universe is an enormous direct product of representations of symmetry groups" - Steve Weinberg

(KLV) Emanuel Knill, Raymond Laflamme, Lorenza Viola, Phys. Rev. Lett. **84**, 2525 (2000)

Collective Decoherence



system of qubits

bath

Bath mode with a ***wavelength much longer than the inner qubit spacing***

Acts on ith qubit

system-bath Hamiltonian

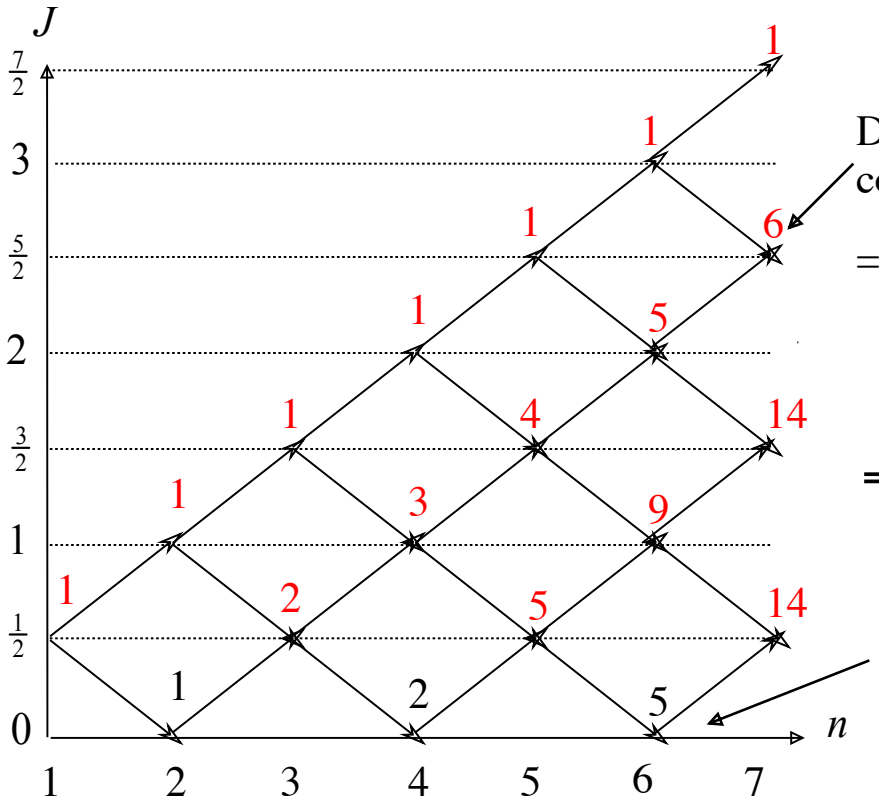
$$H_I = \sum_{i=1}^{n} \sum_{\alpha=\{x,y,z\}} c_{\alpha,i}\, \underbrace{s_\alpha^{(i)}}_{\text{system operator}} \otimes \underbrace{b_{\alpha,i}}_{\text{bath operator}} \Rightarrow_{c_{\alpha,i}=c_\alpha} H_I = \sum_{\alpha=\{x,y,z\}} \sum_{i=1}^{n} c_\alpha s_\alpha^{(i)} \otimes b_\alpha = \sum_{\alpha=\{x,y,z\}} S_\alpha^n \otimes B_\alpha$$

Collective decoherence in quantum optics terms is "**Dicke subradiance**".

Expect collective decoherence in systems of fixed qubits with small inner qubit spacing.

Promising systems: atoms trapped in a cavity, solid state systems at low temperatures, etc.

Collective DFSs



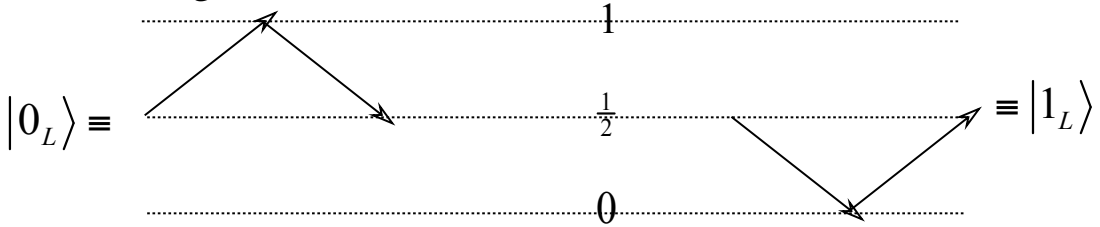
Degeneracy of the n qubit, total angular momentum J collective su(2) operators

= dimension of quantum information which can be encoded into the corresponding DFS.

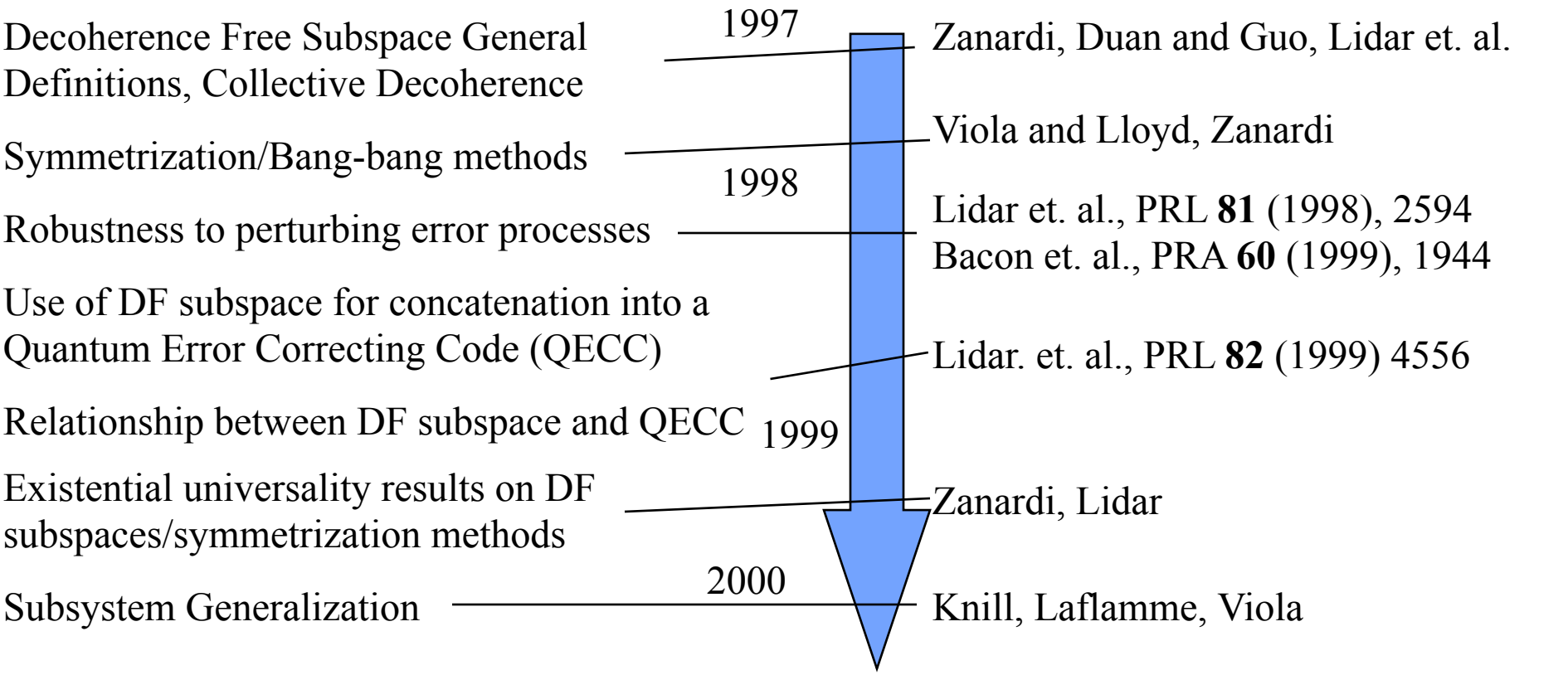
$$= \frac{(2J+1)n!}{\left(\frac{n}{2}+J+1\right)!\left(\frac{n}{2}-J\right)!}$$

The J=0 irreps of su(2) are 1 dimensional Decoherence-Free Subspaces.

A nice set of basis states for a (n,J) DFS can be labeled by the path taken when constructing the (n,J) DFS via addition of angular momentum.



DFS History (Whaley Group Involvement)



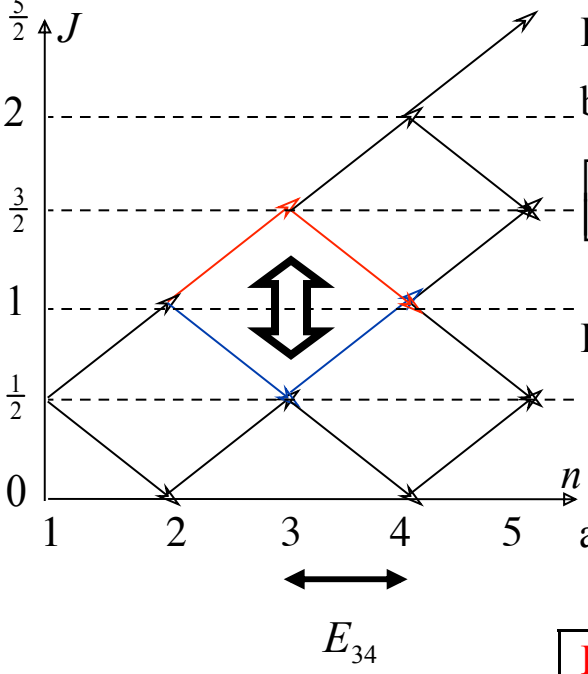
How do we perform quantum computation on the information of a (n,J) DFS?

Can we do this using only one or two-qubit interactions?

Exchange

The Exchange Hamiltonian between 2 qubits $\equiv E = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{matrix} |00\rangle \\ |01\rangle \\ |10\rangle \\ |11\rangle \end{matrix}$. $E_{ij} \equiv$ E acting between qubits i and j.

Exchange commutes with the collective su(2) operators: $\left[E_{ij}, S_{\alpha}^{n}\right] = 0$ if $i, j \le n$



For a fixed number of qubits n (i,j ≤ n), E_{ij} does not mix (n,J) DFSs because E_{ij} commutes with the total angular momentum operator:

$\left[E_{ij}, \left(S^{n}\right)^{2}\right] = 0$ where $\left(S^{n}\right)^{2} = \left(S_{x}^{n}\right)^{2} + \left(S_{y}^{n}\right)^{2} + \left(S_{z}^{n}\right)^{2}$

But, E_{ij} does mix the different basis states!

Think: it changes the order in which the addition of angular momentum is performed.

Exchange is naturally fault tolerant on the DFSs!

The action of the exchange never exposes the info to the DFS decoherence.

The Exchange Hamiltonian between qubits 3 and 4, E_{34}, mixes the red and blue pathway basis states.

Exchange's Universality

Consider the ability to use the exchange Hamiltonian between qubits (nearest neighbors is enough)

$$U_i(t) = \exp\left[\frac{i\lambda E_{i,i+1} t}{\eta}\right] \quad \lambda = \text{coupling constant}, \quad i = 1 \ldots n-1$$

This set of gates is universal on each individual (n,J) DFS! (and not more!)

The exchange Hamiltonian is universal on the encoded states of a collective DFS

Simple example for the $n = 3$, $J = \frac{1}{2}$ DFS:

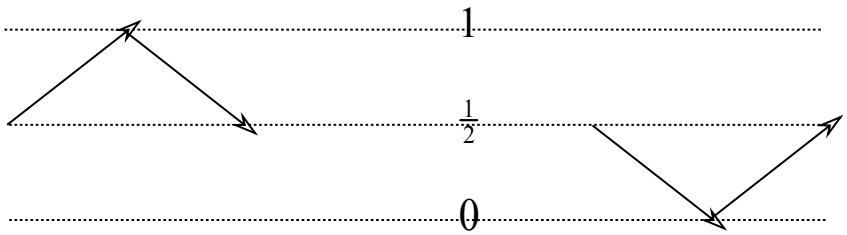
recall that the states are formed from combining a triplet with a spin-$\frac{1}{2}$ state or or by combining a singlet with a spin-$\frac{1}{2}$ state. Triplet is invarient under exchange and singlet flips sign under exchange. Thus

$E_{12} \rightarrow \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$ on the logical qubits.

more details tell us that

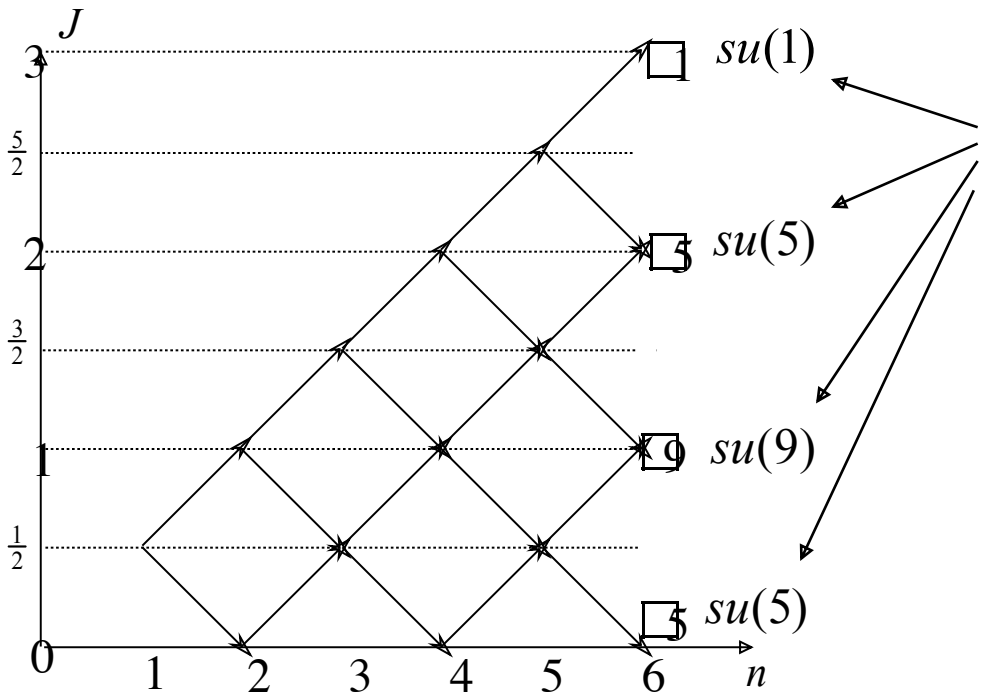
$E_{23} \rightarrow \begin{pmatrix} \frac{1}{2} & \frac{\sqrt{3}}{2} \\ \frac{\sqrt{3}}{2} & -\frac{1}{2} \end{pmatrix}$ on the logical qubits.

$|0_L\rangle \equiv$



$\equiv |1_L\rangle$

Universality Amazing!



Can perform su(d)'s on each of the DFSs

encode information into the DFSs and the EXCHANGE interaction can be used to perform any su(d) on each of the su(d)'s for a given n independently!

su(d) means we have a d dimensional Hilbert space (encoded) and we can perform all unitary manipulations on this d dimensional Hilbert space!

$$\text{encoding efficiency} \equiv r = \frac{\log_2(\text{dimension of DFS})}{n}$$

As $n \to \infty$, $r(n) \to 1 - c\dfrac{\log_2 n}{n}$. The encoding efficiency is nearly unity for large n!

The Exchange Hamiltonian is ASYMPTOTICALLY UNIVERSAL

Heisenberg Coupling

A NEW DESIGN FOR QUANTUM COMPUTER ARCHITECTURE

$H_{heis} = J(t)\, s^{(i)} \cdot s^{(j)}$ where $s^{(i)} = \left(s_x^{(i)}, s_y^{(i)}, s_z^{(i)}\right)$ are the single spin operators on qubit i

$$s^{(i)} \cdot s^{(j)} = s_x^{(i)} s_x^{(j)} + s_y^{(i)} s_y^{(j)} + s_y^{(i)} s_y^{(j)} = \tfrac{1}{4}\left(E_{ij} - I\right)$$

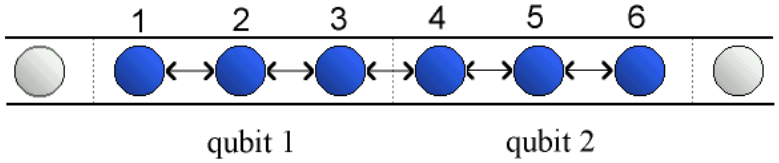
since I in Hamiltonian does not matter (global phase):

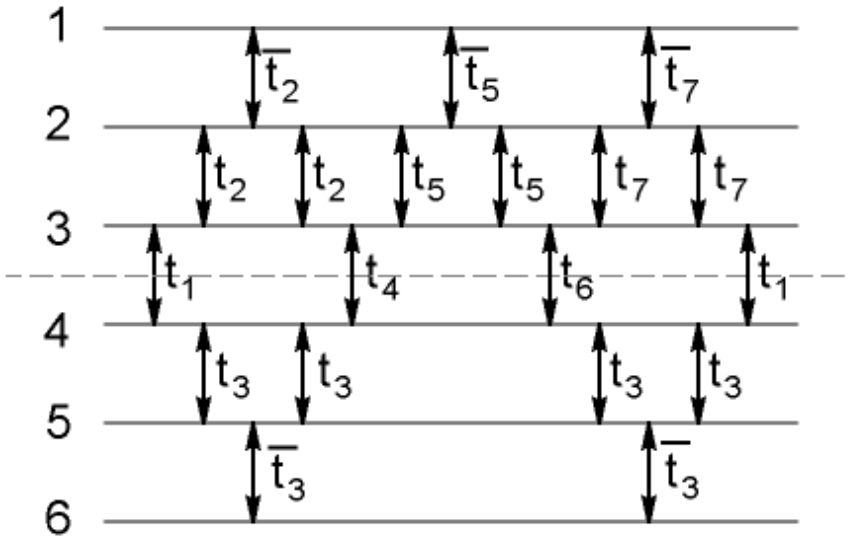
the Heisenberg Coupling is exactly the Exchange Coupling of quantum dot / Kane proposal

Not necessary to supplement the Heisenberg coupling with single qubit operations!

- Heisenberg interaction in *quantum dot / Kane proposal* are easily tuned using local magnetic fields.
- Heisenberg operation is fast (GHz range) and has an on-off ratio of several orders of magnitude.
- Encoding into low J DFSs is easily performed using a simple cooling routine: let there be an exchange interaction between alternating neighboring dots and cool the system.

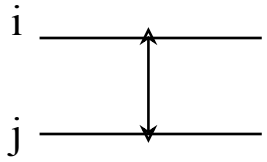
Explicit Gate Construction





t_1=0.410899(2) t_5=0.414720(10)
t_2=0.207110(20) t_6=0.147654(12)
t_3=0.2775258(12) t_7=0.813126(12)
t_4=0.640505(8) $\tan(\pi t)\tan(\pi \bar{t}) = -2$

exchange gate



$= \exp\left(iJE_{i,j}t\right)$

DiVincenzo, Bacon, Kempe, Burkard, Whaley "Universal Quantum Computation with the Exchange Interaction" quant-ph/0005116

What Do I Need To Quantum Compute?



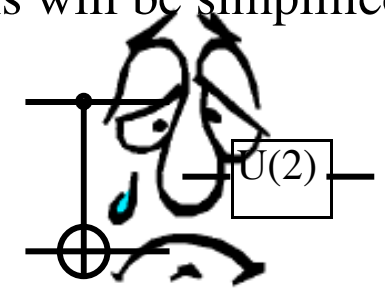
Hey I found this new quantum system with these crazy interactions. Can I quantum compute?

Many interactions, with a bit of encoding, can be used to do quantum computation.

This question is a simple application of the representation theory of Lie Algebras (take it to your local particle theorist for a quick answer!)

A few more qubits and a few more operations may be required.

The hope, however, is that the requirements for engineering a quantum computer using encoded universal interactions will be simplified.



Controlled-Not and single qubit gates are not the only way to go!



Universality and Implementations

Early history of QC: generic set of gates is universal (Lloyd // Deutsch, Barenco, and Ekert)

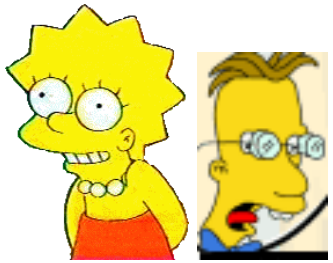
Nature is not generic: **TM**

Nature

DFS Universality Results

Existential Universality	Zanardi, Lidar
Exchange is universal on collective decoherence decoherence-free subspaces	Bacon, Kempe, Lidar, and Whaley, PRL **85** (2000) 1758.
Exchange is universal on collective decoherence decoherence-free subsystems	Kempe, Bacon, Lidar, and Whaley, submitted to PRA, quant-ph/0004064
Explicit exchange gates for use as a universal quantum computer	DiVincenzo, Bacon, Kempe, Burkard, and Whaley, submitted to Nature, quant-ph/0005116
DFS concatenated with QECC robust to "exchange errors".	Lidar, Bacon, Kempe, and Whaley, PRA **61** (2000) 052307
DFS FT computation on multi-qubit DFSs	Lidar, Bacon, Kempe, Whaley, submitted to PRA, quant-ph/9908064 and quant-ph/0007013

Whaley Group Members



Professor K. B. Whaley, Department of Chemistry

Postdoc:

Dr. Daniel Lidar (leaving for faculty of University of Toronto)

DFS theory, quantum dot proposal

Graduate Students:

Dave Bacon, Department of Physics DFS theory, Quantum optics DFSs

Julia Kempe, Department of Mathematics DFS theory, quantum information topics

Ken Brown, Department of Chemistry Quantum dots connected by linear support

Simon Myrgren, Departement of Chemistry Trajectory simulations of exchange universality and quantum dot proposal